Pavlo Fedyna

Senior Software Engineer · Search, AI & Platform Engineering

Hamburg, Germany · Full-time, hybrid or remote

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Summary

Senior backend engineer with 15 years of experience building high-traffic, resilient systems, most recently at OTTO, one of Germany's largest online shops. There I built hybrid lexical and semantic search over 20 million products and an LLM-based pre-sales shopping assistant, and I now work on the product detail page, which handles about half of otto.de traffic. I own services from design to on-call with a strong focus on observability and performance, and help shape OTTO's developer platform as a committee member. I've led a small team, mentor junior engineers, and work best with XP practices like TDD, pairing and continuous delivery.

Skills

Languages	Kotlin, Java, Rust, Go, Python, TypeScript, SQL
Search & AI	Solr, Hybrid search, Semantic search, LLM agents, LangChain4j
Backend	Spring, Spring WebFlux, Helidon, Kafka, Hazelcast, GraphQL
Data	Amazon Aurora, DynamoDB, PostgreSQL, Spark, Airflow, Databricks
Cloud & DevOps	AWS, AWS ECS, AWS CDK, Terraform, nginx, Docker, Kubernetes, GitHub Actions, Jenkins
Observability	OpenTelemetry, Grafana, Prometheus, Mimir, Langfuse
Frontend	React, Svelte
Practices	XP, TDD, Pair programming, CI/CD, A/B testing, On-call, ADRs, Platform engineering

Experience

Senior Software Engineer, Product Detail Page · OTTO (GmbH & Co KG) *02/2026 – present*

- Develop the product detail page for otto.de, backend and frontend, which serves about half of the shop's traffic (around 500 requests/sec).
- Own observability and performance for the team, from metrics and alerting to finding latency bottlenecks.
- Member of OTTO's developer platform committee. Build and advocate for standard infrastructure modules (CI/CD pipelines, Terraform) and am preparing the team to pilot the new platform.
- Built and open-sourced otto-config (github.com/otto-de/otto-config), a shared Java library for reading configuration from multiple sources, now used by the search and product detail page teams.
- Write ADRs for the team and across the department, and validate changes through A/B tests before rolling them out.
- Mentor junior developers, working students and apprentices.

Tech: Kotlin, Go, TypeScript, Svelte, AWS ECS, Amazon Aurora, DynamoDB, Kafka

Senior Software Engineer, Search · OTTO (GmbH & Co KG) *10/2021 – 01/2026*

- Designed and built the search platform for otto.de, with a Solr cluster holding 20 million active products and serving around 200 requests/sec.
- Built and ran the indexing pipeline, including lexical analysis, which keeps the search index within 15 minutes of product data changes.
- Drove Rust adoption in the team and rewrote the hybrid (lexical and semantic) search service in Rust, cutting latency for comparable requests from 100 ms to 20 ms and halving compute.
- Built an LLM-backed query intent service that interprets search queries.
- Owned observability and resilience for the team's services and reduced AWS running costs.
- Migrated infrastructure from CloudFormation to Terraform and CI/CD from Jenkins to GitHub Actions.

Pre-sales AI shopping assistant

- Built an agentic LLM assistant that answers pre-sales questions for up to 100 concurrent users.
- Set up its observability with OpenTelemetry, Langfuse, Grafana and the Prometheus stack.
- Developed the infrastructure for evaluating and measuring the quality of its answers.
- Prototyped voice-to-voice conversational search on top of it.

Live traffic shadowing

- Built a Rust proxy that, together with nginx request mirroring, shadows live production traffic into development environments. It works for any service in the stack.
- Made the share of mirrored traffic adjustable on the fly through reloadable nginx config.
- Used across the team and partly across the department. Besides scaling and infrastructure issues, it caught logic defects that tests had missed.

Tech: Java, Kotlin, Rust, Python, Spring WebFlux, Helidon, Solr, Kafka, Hazelcast, LangChain4j, Langfuse, nginx, AWS, Terraform

Tech Lead · Cargonexx *08/2020 – 09/2021*

- Promoted from Senior Software Engineer to Tech Lead after three months; led a team of 5 engineers.
- Designed and built a SaaS product for fleet and shipment management (Java/Spring, React).
- Built a tour cost and time optimization service; integrated GPS providers for live fleet tracking.
- Set quarterly OKRs with the product owner and refined requirements before they went into development.
- Introduced TDD, pair programming and CI/CD to the team.

Tech: Java, Spring, TypeScript, React, PostgreSQL, AWS, Docker

Senior Software Engineer · Zalando Marketing Services *02/2018 – 08/2020*

- Built the reporting for Zalando's self-service B2B advertising platform.
- Rewrote the analytics pipeline and moved it to the cloud, cutting daily operating costs by 10x.
- Built high-load consumers that process Zalando's full view and click stream for analytics.
- Built a marketplace analytics module and self-service data access for stakeholders.
- Set up 24/7 on-call for the team.

Tech: Python, Java, Scala, Spark, Airflow, Databricks, PostgreSQL, GraphQL, AWS, Kubernetes

Senior Software Engineer · EOS Technology Solutions *08/2017 – 02/2018*

- Refactored and tested a legacy client data system, tuned the ORM for a faster UI, and introduced BDD.

Tech: Java, Spring, Hibernate, React, Kafka, Cucumber, Selenium

Lead Software Engineer · Smaato *10/2014 – 08/2017*

- Joined as Software Engineer and was promoted to Lead Software Engineer in 2017.
- Reworked the auction algorithm of a distributed ad auction system handling about 30,000 requests/sec on average, cutting CPU usage by 20% and allowing a smaller cluster.
- Built a plugin framework so sales engineers could onboard advertisers without developer help.
- Led development of the B2B client portal and delivered the MVP on schedule with a short-staffed team.
- Migrated and rewrote several legacy applications.

Tech: Java, Spring, Hibernate, Kafka, React, AWS, Docker, Packer

Software Engineer · Edvantis, Lvivsoft *07/2011 – 09/2014*

- Java/Spring development for clients in the US and Europe, including a social media platform.

Tech: Java, Spring, SQL, Solr, AWS

Education

Bachelor's degree, Applied Linguistics · Lviv Polytechnic National University *2006 – 2010*

Languages

English (C2), German (C1), Ukrainian (Native)

Interests

- **Embedded Rust** — Bare-metal and IoT projects on microcontrollers.
- **Microelectronics** — Hands-on electronics and hardware projects.
- **Homelab** — Proxmox cluster running a fully functional Kubernetes setup for learning and experimentation.
- **Game development** — Small games as a way to explore real-time systems.